

李庭競 Lee Ting Ching

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EXPERIENCE

AUO Corporation Sept. 2024 – Present
Software Engineer (RDSS) Taoyuan, TW

- Design and optimize a routing algorithm to boost the SLT equipment's takt time by approximately 30%.
- Develop a workflow that helps reduce waste materials by 10% and improve product precision by 50% in nanometers via ML.
- Construct a novel algorithm to address uneven incoming materials for uLED panels, capable of completing calculations within 1 minute.
- Deploy a backend system by Django to integrate with MES, supporting the above three aspects in the mass production environment.
- Utilize the vibe coding and AI agent skills to rapidly respond to users' emergency requests.

TAAI Mar. 2023 – Sept. 2024
Network Administrator Hsinchu, TW

- Assist with website maintenance, front/back-end integration development for the credit-card system, and setting up a forum website.
- Built the website using Vue.js, managed the payment database with PostgreSQL, and handled member data with MongoDB.
- Deployed to Heroku with managed DNS and SSL.

IRISSatelite, NCKU Apr. 2021 – Dec. 2021
Ground Station Software Developer Tainan, TW

- Developed satellite image integration and backend parsing functions.
- Used C# for MVC architecture refactoring and development, utilizing basic OOP principles to manage user-defined data.

PROJECTS

Synchronous Laser Routing Strategies Jan. 2023 – Dec. 2023
Owner and researcher SYNTEC/NTHU

- Formulated a constrained TSP evolutionary algorithm for laser route planning on CNC machines, achieving shorter total length and smoother velocity.
- Parallelized programming to accelerate the evolution time.

Minimum PCB Area Layout via Evolution Algorithm Aug. 2022 – Mar. 2024
Constraint designer AcBel/NTHU

- Formulated a constrained multi-objective evolutionary algorithm for PCB Area Layout, achieving smaller area and shorter length.
- Using C++ with OOP to develop large-scale projects, utilizing CMakeLists for multi-threaded compilation.
 - Managing collaborative version control with Git, and applying basic design patterns for architectural design.

EDUCATION

National Tsing Hua University Aug. 2022 – Aug. 2024
Master of Power Mechanical Engineering and Artificial Intelligence team Hsinchu, TW

Synchronous Laser Routing Strategies Based on Evolutionary Algorithm

- IEEE Conference on Artificial Intelligence 2024
- Formulated a constrained TSP evolutionary algorithm for laser route planning on CNC machines, achieving shorter total length and smoother velocity.
- Program skills: Parallelized programming, Optimization Memetic Algorithm
- Coursework: Data Mining, Algorithms, Numerical Optimization, Artificial Intelligence, Evolution Algorithm

National Cheng Kung University Aug. 2018 – Jun. 2022
Bachelor of Mechanical Engineering Tainan, TW

NCKU Robotic Project in Arduino

- Led the design of a robot with Bluetooth remote control.
- Project skills: Multi-task control of servos, steppers, and DC motors
- Coursework: Object-Oriented Programming, Computer Vision, Computing Theory

SKILLS

Languages: Python, C++, C#, HTML/CSS/JS

Tools: Google OR tools, Nvidia cuOpt, Docker, Git, Swagger(RESTful), MongoDB, PostgreSQL, Linux main., GCP, Azure

AI skills: Codex, langflow, agent skills, roo coder(EOL)

Here is my development note: <https://dev.crazyryan.xyz/>